

HMMT February 2015
Saturday 21 February 2015

HMIC
Testing window: April 1–5

1. [20] Let S be the set of positive integers n such that the inequality

$$\phi(n) \cdot \tau(n) \geq \sqrt{\frac{n^3}{3}}$$

holds, where $\phi(n)$ is the number of positive integers $k \leq n$ that are relatively prime to n , and $\tau(n)$ is the number of positive divisors of n . Prove that S is finite.

Answer: N/A

Proof. Let S be the set of all positive integers n such that

$$\phi(n) \cdot \tau(n) \geq \sqrt{\frac{n^3}{3}}.$$

Define a function Φ on all positive integers n by

$$\Phi(n) = \frac{\phi(n)^2 \cdot \tau(n)^2}{n^3}.$$

An important observation is that Φ has the property that for every relatively prime positive integers m, n , we have $\Phi(mn) = \Phi(m)\Phi(n)$.

Define another function ψ on all ordered-pairs (a, p) of positive integer a and prime number p as follows:

$$\psi(a, p) := \frac{(a+1)^2(1-1/p)^2}{p^a}.$$

If we express n in its canonical form as $n = \prod_{i=1}^k p_i^{a_i}$, then we have

$$\Phi(n) = \prod_{i=1}^k \psi(a_i, p_i).$$

Therefore, S is actually the set of all $n = \prod_{i=1}^k p_i^{a_i}$ such that

$$\prod_{i=1}^k \psi(a_i, p_i) \geq \frac{1}{3}.$$

It is straightforward to establish the following: for every prime p and positive integer a ,

- if $p \geq 11$, then $\psi(a, p) < \frac{1}{3}$;
- if $p = 5$ and $a \geq 2$, then $\psi(a, p) < \frac{1}{3}$;
- if $p = 3$ and $a \geq 3$, then $\psi(a, p) < \frac{1}{3}$;
- if $p = 2$ and $a \geq 5$, then $\psi(a, p) < \frac{1}{3}$;
- $\psi(a, p)$ is always less than $\frac{1}{\sqrt{3}}$ **unless** $(a, p) = (1, 3)$ where $\psi(1, 3) = \frac{16}{27}$.

The data above shows that in the case $n = p^a$, in which there is only one prime dividing n , there are at most 8 possible n in S : $2^1, 2^2, 2^3, 2^4, 3^1, 3^2, 5^1, 7^1$. If n is divisible by at least two distinct primes,

then one of them must be 3^1 and 3^1 fully divides n (that is, $3^2 \nmid n$). In the latter case $n = 3 \cdot n_0$. Write $n_0 = \prod_{j=1}^l q_j^{b_j}$. In order for $\Phi(n) \geq \frac{1}{3}$, we require

$$\prod_{j=1}^l \psi(b_j, q_j) \geq \frac{9}{16}.$$

This is only possible when $(b, q) = (2, 2)$, where $n = 12$. Also, note that when $n = 1$, $\Phi(1) = 1$, so $1 \in S$. Hence, there are at most 10 (a finite number of) possible values of n in S . $\square$

Remark 0.1. This problem was proposed by Pakawut Jiradilok.

Remark 0.2. By explicit calculations, there are in fact exactly ten elements in S : 1, 2, 3, 4, 5, 7, 8, 9, 12, 16.

Remark 0.3. To only get a finite bound, one does not need to be nearly as careful as we are in the above proof.

2. [25] Let m, n be positive integers with $m \geq n$. Let S be the set of pairs (a, b) of relatively prime positive integers such that $a, b \leq m$ and $a + b > m$.

For each pair $(a, b) \in S$, consider the nonnegative integer solution (u, v) to the equation $au - bv = n$ chosen with $v \geq 0$ minimal, and let $I(a, b)$ denote the (open) interval $(v/a, u/b)$.

Prove that $I(a, b) \subseteq (0, 1)$ for every $(a, b) \in S$, and that any fixed irrational number $\alpha \in (0, 1)$ lies in $I(a, b)$ for exactly n distinct pairs $(a, b) \in S$.

Answer: N/A

Proof. The fact that $I(a, b) \subseteq (0, 1)$ follows from the small-ness of $n \leq m < a + b$: the smallest solution (u, v) has $0 \leq v \leq a - 1$, so $u = \frac{n+bv}{a} < \frac{(a+b)+b(a-1)}{a} = b + 1$ forces $1 \leq u \leq b$.

For the main part of the problem, it suffices (actually, is equivalent) to show that

- (i) 0, 1 each appear exactly n times as endpoints of intervals $I(a, b)$; and
- (ii) each reduced rational $p/q \in (0, 1)$ appears an equal (possibly zero, if q is large) number of times as left and right endpoints of intervals $I(a, b)$.

We prove these separately as follows:

- (i) 0 is a (left) endpoint precisely when $v/a = 0$, or equivalently $(a, b) \in S$ has $a \mid n$ (look at $v \bmod a$). Since $n \leq m$, we get $\phi(d)$ good pairs $(d, b) \in S$ for fixed $d \mid n$. Indeed, for any coprime residue class modulo d , there's exactly one representative b with $b \leq m < b + d$. (Conceptually, it may be enlightening to think of S as part of the Euclidean algorithm tree generated by $(1, 1)$.)

Thus 0 occurs as an endpoint $\sum_{d \mid n} \phi(d) = n$ times. Similarly, 1 is an endpoint precisely when $u = b$, or equivalently $(a, b) \in S$ has $b \mid n$ (look at $u \bmod b$). By the same reasoning as before, we get n right endpoint occurrences of 1.

- (ii) Fix $p/q \in (0, 1)$ with p, q coprime, so $q \geq 2$. We want to show that $v_1/a_1 = p/q$ occurs for the same number of $(a_1, b_1) \in S$ as $u_2/b_2 = p/q$ occurs for $(a_2, b_2) \in S$. (In fact, we will show that the number of occurrences of the former in $(x, b_1) \in S$ equals the number of the latter in $(a_2, x) \in S$.) The former occurs precisely when $1 \leq a_1 \leq m$, $q \mid a_1$, and (given the first two conditions) $v_1 \equiv p(a_1/q) \pmod{a_1}$. Given the first two conditions, the third is equivalent to $-nb_1^{-1} \equiv p(a_1/q) \pmod{a_1}$, or $b_1(a_1/q) \equiv -np^{-1} \pmod{a_1}$. This is equivalent to having $(a_1/q) \mid n$ and $b_1 \equiv (-n/(a_1/q))p^{-1} \pmod{q}$.

Similarly, the latter occurs precisely when $1 \leq b_2 \leq m$, $q \mid b_2$, $(b_2/q) \mid n$, and $a_2 \equiv (n/(b_2/q))p^{-1} \pmod{q}$.

As hinted at before, if we consider the occurrences for a fixed value $x = a_1 = b_2$, then the number of permitted residue classes $b_1 \pmod{a_1 = x}$ (in the former) is the same as the number of permitted residue classes $a_2 \pmod{b_2 = x}$ (in the latter): $y \pmod{x}$ is permitted in the former if and only if $-y \pmod{x}$ is permitted in the latter; note that $(y, x) = 1$ if and only if $(-y, x) = 1$.

□

Remark 0.4. This problem was proposed by Victor Wang.

Remark 0.5. This problem was inspired by ISL 2013 N7. Note that the $n = 1$ case gives the famous Farey fractions sequence (for fractions of denominator at most m).

3. [30] Let M be a 2014×2014 invertible matrix, and let $\mathcal{F}(M)$ denote the set of matrices whose rows are a permutation of the rows of M . Find the number of matrices $F \in \mathcal{F}(M)$ such that $\det(M + F) \neq 0$.

Answer: 2013!!²

Proof. Number the rows of M as $r_1, r_2, \dots, r_{2014}$, and remark that by invertibility, these rows are linearly independent. We can write $\det(M + F)$ as the sum of the determinants of the 2^{2014} matrices which can be obtained by replacing some set of the rows of M with the corresponding rows of F (this follows from distributivity and expansion by minors) - we'll call these the insertion matrices of M . Of course, the rows of F are themselves some permutation of the rows of M .

Now, all insertion matrices in which some row appears at least twice have determinant 0 (by linear dependence of rows), so the determinant of $M + F$ is the sum of the determinants of the insertion matrices whose rows are a permutation of the r_i . Let the i -th row of F be $r_{\sigma(i)}$ and partition σ into disjoint cycles; we claim $\det(M + F) = 0$ iff σ contains an even cycle.

Indeed, if we replace row r_i in M , in order for the rows of $M + F$ to remain a permutation of the r_i we must replace row $r_{\sigma(i)}$ as well, and so on. In other words, we either replace none of the rows in any particular cycle, or we replace all of them. Yet if any cycle is even, the determinants of any insertion matrix in which that cycle is replaced and the same insertion matrix, except not replacing that cycle, are the same in magnitude and opposite in sign (this follows from the fact that a row transposition flips the sign of the determinant, and we must perform an odd number of these). Thus $\det(M + F)$ vanishes if σ contains any even cycles. On the other hand, if σ consists of only odd cycles, every non-zero insertion matrix has determinant $\det(M) \neq 0$ by analogous reasoning (this time, the even number of row transpositions preserves sign). It remains only to count the number of elements of S_{2014} which can be decomposed into exclusively odd cycles.

We begin by writing a recursion for the general problem. Let x_n denote the number of permutations on $[n]$ which can be decomposed into exclusively odd cycles. By casework on the cycle containing 1, we find the recursion

$$x_n = \sum_i \binom{n-1}{2i-2} (2i-2)! x_{n-2i+1}$$

where $x_0 = x_1 = x_2 = 1$.

The intended solution from this point was combinatorial in nature, similar to the one from the fourth comment under Evan Chen's "Recursion" blog post (from June 2012). The following much nicer finish is due to Lawrence Sun:

Set $y_n = x_n/n!$ and rewrite the recursion as

$$ny_n = y_{n-1} + y_{n-3} + \dots$$

Shifting, we find in addition that

$$(n-2)y_{n-2} = y_{n-3} + \dots$$

Subtracting,

$$ny_n = y_{n-1} + (n-2)y_{n-2}$$

for all n . It now follows from a straightforward induction that $y_{2k} = [(2k-1)!!]^2/(2k)!$ and $y_{2k+1} = [(2k+1)!!(2k-1)!!]/(2k+1)!$. In particular,

$$x_{2014} = 2014!y_{2014} = (2013!!)^2$$

as desired.

Here is an alternative approach: define the power series $F(z)$ in the variable z by

$$F(z) = \sum_{n=0}^{\infty} x_n \frac{z^n}{n!}.$$

The recurrence relation is equivalent to the differential equation

$$F'(z) = \left(\sum_{n=0}^{\infty} z^{2n} \right) F(z) = \frac{1}{1-z^2} F(z).$$

Solving this equation using the initial condition $F(0) = 1$ gives

$$F(z) = \sqrt{\frac{1+z}{1-z}} = (1+z)(1-z^2)^{-1/2}.$$

By the binomial theorem, we have

$$F(z) = (1+z) \sum_{n=0}^{\infty} \binom{-1/2}{n} (-1)^n z^{2n},$$

so

$$x_{2014} = 2014! \binom{-1/2}{1007} (-1)^{1007} = (2013!!)^2$$

as desired. □

Remark 0.6. This problem was proposed by Dhroova Aiyam.

Remark 0.7. We can write the matrices $F \in \mathcal{F}(M)$ as $F = PM$, where P ranges over all 2014×2014 permutation matrices. Then, we have $F + M = (P + 1_{2014 \times 2014})M$, which is invertible if and only if $P + 1_{2014 \times 2014}$ is. Using this, we can give an alternative proof that the answer to the question is equal to the number of permutations of $\{1, \dots, 2014\}$ with cycles of odd length.

4. [35] Prove that there exists a positive integer N such that for any positive integer $n \geq N$, there are at least 2015 non-empty subsets S of $\{n^2 + 1, n^2 + 2, \dots, n^2 + 3n\}$ with the property that the product of the elements of S is a perfect square.

Answer: N/A

Lemma 0.1. Let n, k be integers with $1 \leq k \leq n$. Then

$$\binom{n}{k} \leq \left(\frac{en}{k} \right)^k$$

where $e \approx 2.71828 \dots$ is Euler's number.

Proof. We have

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k!} \\ &\leq \frac{n^k}{k!} \\ &= \left(\frac{n}{k} \right)^k \frac{k^k}{k!} \\ &\leq \left(\frac{n}{k} \right)^k \sum_{m=0}^{\infty} \frac{k^m}{m!} \\ &= \left(\frac{en}{k} \right)^k \end{aligned}$$

where the last step follows from the series expansion

$$e^x = \sum_{m=0}^{\infty} \frac{x^m}{m!}$$

applied to $x = k$. □

In the solution below, all logarithms are to the base e . If f and g are two functions of the variable n , then write $f(n) \sim g(n)$ to denote that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1.$$

Proof of original problem. Let U be the set of elements of $\{n^2 + 1, \dots, n^2 + 3n\}$ with no prime factor greater than $3n$.¹ Let s be the number of primes $p \leq 3n$. By the pigeonhole principle, we can find distinct subsets $X_1, \dots, X_r \subset U$ with $r \geq \frac{2^{|U|}}{2^s}$ such that every prime $p \leq 3n$ either divides all the products

$$\prod_{x \in X_1} x, \dots, \prod_{x \in X_r} x$$

an even number of times or divides all of them an odd number of times. Define the subsets $Y_1, \dots, Y_r \subset U$ by

$$Y_i := X_1 \Delta X_i = (X_1 - X_i) \cup (X_i - X_1)$$

for $1 \leq i \leq r$. Then by construction, the sets $Y_1, \dots, Y_r$ are distinct, and they have the property that every prime $p \leq 3n$ divides each of the products

$$\prod_{y \in Y_1} y, \dots, \prod_{y \in Y_r} y$$

an even number of times. Since $Y_1, \dots, Y_r \subset U$, the products

$$\prod_{y \in Y_1} y, \dots, \prod_{y \in Y_r} y$$

do not have any prime factor greater than $3n$, so they are perfect squares.

We have found r distinct subsets of $\{n^2 + 1, \dots, n^2 + 3n\}$ whose products are perfect squares. Therefore, it remains only to show that $r \geq 2016$ for n large enough. Since $r \geq \frac{2^{|U|}}{2^s}$, it suffices to show that $|U| \geq s + 11$ for n large enough.

Let T be the set of prime numbers $p > 3n$ that divide an element of $\{n^2 + 1, \dots, n^2 + 3n\}$. Every element of T evidently divides the binomial coefficient

$$\binom{n^2 + 3n}{3n} = \frac{(n^2 + 1) \cdots (n^2 + 3n)}{(3n)!},$$

so the product of the elements of T divides $\binom{n^2 + 3n}{3n}$ as well. We then have the chain of inequalities

$$\begin{aligned} (3n)^{|T|} &\leq \prod_{p \in T} p \\ &\leq \binom{n^2 + 3n}{3n} \\ &\leq \left(\frac{e(n+3)}{3} \right)^{3n} \end{aligned}$$

¹The reason for choosing the threshold $3n$ will become clear.

where the last inequality follows from the lemma. Therefore,

$$\begin{aligned} |T| &\leq \frac{\log \left(\left(\frac{e(n+3)}{3} \right)^{3n} \right)}{\log(3n)} \\ &= 3n \frac{1 + \log(n+3) - \log 3}{\log 3 + \log n}. \end{aligned}$$

It follows that

$$\begin{aligned} |U| &\geq 3n - |T| \\ &\geq 3n \left(1 - \frac{1 + \log(n+3) - \log 3}{\log 3 + \log n} \right) \\ &\sim (2 \log 3 - 1) \frac{3n}{\log(3n)}. \end{aligned}$$

On the other hand, by the prime number theorem we have

$$s \sim \frac{3n}{\log(3n)}.$$

Since $2 \log 3 - 1 > 1$, we conclude that $|U| \geq s + 11$ for n large enough, as desired. $\square$

Remark 0.8. This problem was proposed by Dhroova Aiyilam. Thanks to Mitchell Lee for writing the solution, and also working out Remark 0.10.

Remark 0.9. In fact, the above proof gives a lower bound for r of the form $2^{(2 \log 3 - 2 + o(1)) \frac{3n}{\log(3n)}}$ where $o(1)$ denotes a term that approaches 0 as $n \rightarrow \infty$.

Remark 0.10. In fact, for any real number $\epsilon > 0$ there is an integer N_ϵ such that the set

$$\left\{ n^2 + 1, \dots, n^2 + \left\lceil \left(\sqrt{\frac{3}{2}} e^{3/2} + \epsilon \right) \frac{n}{\log n} \right\rceil \right\}$$

contains a nonempty subset whose product is a square for any $n > N_\epsilon$. The proof of this is similar to the above solution with $3n$ replaced by

$$\left(\sqrt{\frac{3}{2}} e^{3/2} + \epsilon \right) \frac{n}{\log n}$$

and a more careful lower bound for

$$\prod_{p \in T} p.$$

Remark 0.11. Compare with the following problem (addressing higher powers): http://www.aops.com/community/c6h495623_perfect_powers_in_intervals.

5. [40] Let $\omega = e^{2\pi i/5}$ be a primitive fifth root of unity. Prove that there do not exist integers a, b, c, d, k with $k > 1$ such that

$$(a + b\omega + c\omega^2 + d\omega^3)^k = 1 + \omega.$$

Answer: N/A

Proof. Let $\zeta = \omega = e^{2\pi i/5}$ be a primitive fifth root of unity. If $x = a + b\zeta + c\zeta^2 + d\zeta^3$ ($a, b, c, d \in \mathbb{Z}$), then from $\zeta + \zeta^{-1} = \frac{\sqrt{5}-1}{2}$ and $\zeta^2 + \zeta^{-2} = \zeta^3 + \zeta^{-3} = -\frac{\sqrt{5}+1}{2}$, we deduce $|x|^2 = (a^2 + b^2 + c^2 + d^2) + \frac{\sqrt{5}-1}{2}(ab + bc + cd) - \frac{\sqrt{5}+1}{2}(ac + bd + ad) \in \mathbb{Z}[\phi]$, where $\phi = \frac{1+\sqrt{5}}{2}$.

But $|x|^{2k} = |1 + \zeta|^2 = 2 + \phi^{-1} = \phi^2$, so $\phi^{2/k} = |x|^2 \in \mathbb{Z}[\phi]$. It follows that $|x|^2$ has norm $\phi^{2/k} \bar{\phi}^{2/k}$ in $\mathbb{Z}[\phi]$, and so must be a positive unit. Yet by the lemma below, the positive units are precisely the integer powers of ϕ , so $k > 1$ forces $k = 2$.

However, there must exist $f, g \in \mathbb{Z}[t]$ such that $f(t)^k = 1 + t + g(t)(t^4 + t^3 + t^2 + t + 1)$ (if $f(\zeta) = x$, then the minimal polynomial of ζ over $\mathbb{Q}$ must divide $f(t)^k - 1 - t$), whence $f(1)^k \equiv 2 \pmod{5}$, so $2 \nmid k$, contradiction.

(Alternatively, $k = 2$ would force $a^2 + b^2 + c^2 + d^2 - ab - bc - cd = 0$, which can only occur if $a = b = c = d = 0$.²) $\square$

Lemma 0.2. *The positive units of $\mathbb{Z}[\phi]$ are precisely the integer powers of ϕ . (In other words,³ ϕ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$.)*

Proof. To prove this, one can either use Pell equation theory, or reason more directly as follows: a unit $m + n\phi > 1$ must be at least ϕ . (Sketch: unit is equivalent to norm ± 1 , and so its conjugate $m + n\bar{\phi} = \frac{\pm 1}{m + n\phi}$ lies in $(-1, 1)$, so $n\sqrt{5} = n(\phi - \bar{\phi}) > 0$ forces n positive, and then $-1 < m + n\bar{\phi}$ forces $m > -n\bar{\phi} - 1 = \frac{n}{\phi} - 1 > -1$. Hence $m + n\phi \geq 0 + 1\phi = \phi$, as desired.) If it's not a positive power of ϕ , then for some $\ell > 0$, $\frac{m + n\phi}{\phi^\ell}$ is a unit lying in $(1, \phi)$, contradiction. $\square$

Remark 0.12. This problem was proposed by Carl Lian.

Remark 0.13. The “morally correct” (purely algebraic) way to think about the above solution is via the tower of (number field) extensions

$$\mathbb{Q}(\zeta)/\mathbb{Q}(\sqrt{5})/\mathbb{Q}.$$

Instead of thinking in terms of complex absolute value $|\cdot|$, it is better to reason in terms of the norm $N_{\mathbb{Q}(\zeta)/\mathbb{Q}(\sqrt{5})}(\cdot): \mathbb{Q}(\zeta) \rightarrow \mathbb{Q}(\sqrt{5})$ (which happens to be the same as looking at the square of complex absolute value, in our case), which maps algebraic integers to algebraic integers; the norm of $x^k = 1 + \zeta$ over $\mathbb{Q}(\sqrt{5})$ is ϕ^2 , and ϕ is the *fundamental unit* of $\mathbb{Q}(\sqrt{5})$. Then one proceeds as in the above solution.

²On a silly note, this inspired HMMT November 2013 General Problem 7.

³since $\mathbb{Z}[\phi]$ is the *ring of integers* of $\mathbb{Q}(\sqrt{5})$